

Dr. Ubeiden C. Samboni
Facultad de Ciencias Químicas
Universidad Nacional de Córdoba
Ciudad Universitaria, Córdoba
X5000HUA, Argentina
<https://www.fcq.unc.edu.ar>

Email: ubeiden-samboni@unc.edu.ar

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Professor Dame Wendy Hall
Editor-in-Chief
Royal Society Open Science

I am pleased to submit our manuscript entitled “*Theoretical insights into the electronic structure and NBO features driving nonlinear optical enhancement in the donor- π -acceptor system of OPPh₃py*” for consideration as a research article in *Royal Society Open Science*.

This study presents a detailed DFT and NBO investigation of diphenyl [2-(2-pyridylaminomethyl)phenyl]phosphine oxide (OPPh₃py), focusing on the structural and electronic factors governing its nonlinear optical (NLO) performance. Optimized geometries reproduce experimental parameters and reveal an intramolecular N-H...O hydrogen bond that enhances electronic delocalization. Frontier molecular orbital and NBO analyses confirm a strong donor- π -acceptor framework enabling efficient intramolecular charge transfer. Notably, OPPh₃py exhibits high polarizability and significantly enhanced first- (β) and second-order (γ) hyperpolarizabilities, outperforming *p*-nitroaniline under near-infrared excitation. These findings highlight its promise for photonic and optoelectronic applications and provide insight for the rational design of advanced NLO materials.

This manuscript is original, unpublished, and not under consideration elsewhere. We believe it will be of strong interest to the readership of *Royal Society Open Science*.

Thank you for considering our submission. We look forward to your positive response.

Sincerely,

Dr. Ubeiden C. Samboni